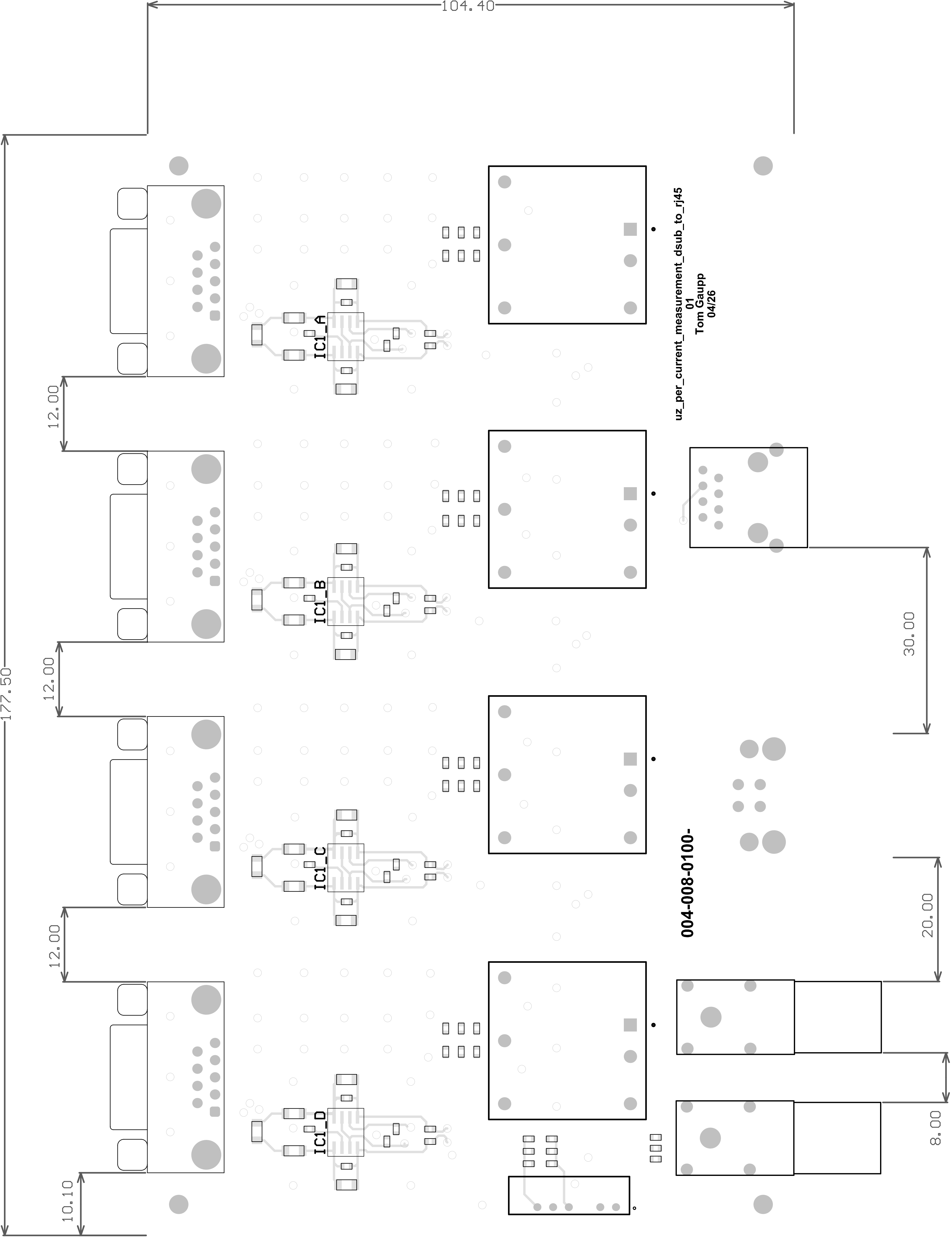
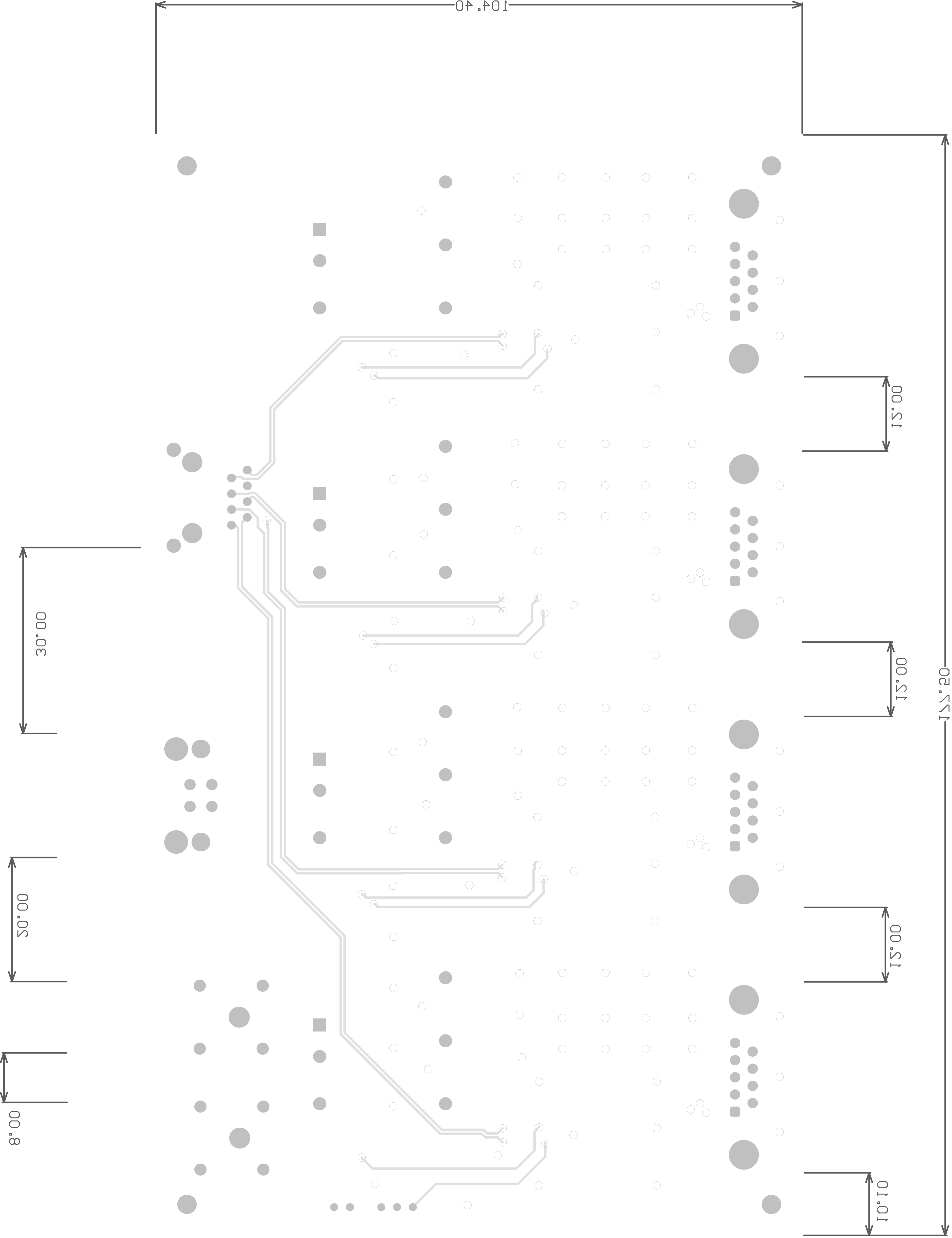




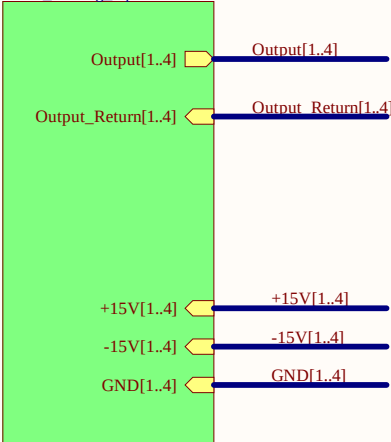


TopSheet

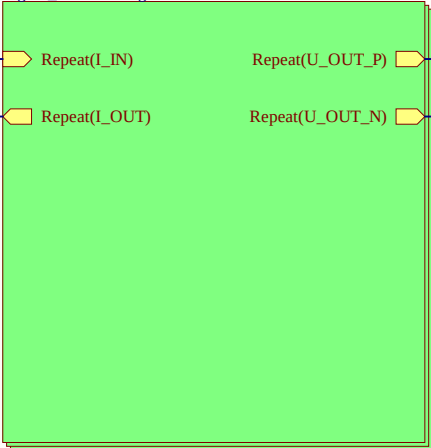
Output1	I_IN_AMP1
Output_Return1	I_OUT_AMP1
Output2	I_IN_AMP2
Output_Return2	I_OUT_AMP2
Output3	I_IN_AMP3
Output_Return3	I_OUT_AMP3
Output4	I_IN_AMP4
Output_Return4	I_OUT_AMP4

U_OUT_P1	A_OUT_P1
U_OUT_N1	A_OUT_N1
U_OUT_P2	A_OUT_P2
U_OUT_N2	A_OUT_N2
U_OUT_P3	A_OUT_P3
U_OUT_N3	A_OUT_N3
U_OUT_P4	A_OUT_P4
U_OUT_N4	A_OUT_N4

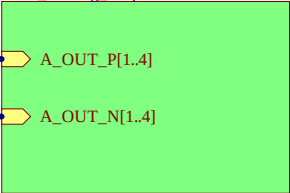
DSub_Input
DSub_Analog_Input



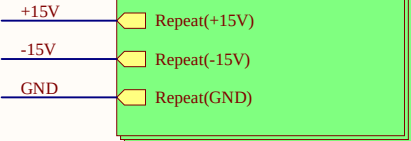
REPEAT(Signal_Conditioning,1,4)
Signal_Conditioning



RJ45_Output
RJ45_Analog_Output



REPEAT(PS_CT,1,4)
Power_Supply_Current_Transformers



PS_Signal_Conditioning
Power_Supply_Signal_Conditioning



Chose size of UZ logo
by chosing footprint
type



INFO1

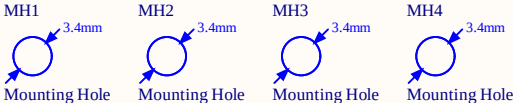
ZC_GGG
ZC_PPP
ProjectRevision
ZC_vv

ZC Serialnumber

INFO2

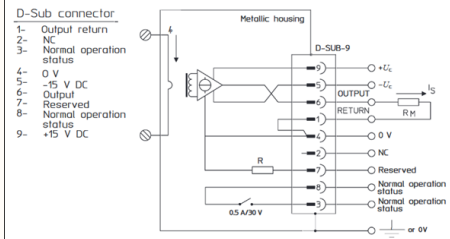
Project
ProjectRevision
AuthorParam
ProjectDate

Design Information



Place MH close to edge connector
when ordering PCB (without
assembly) from JLC. They ensure
accurate outline milling of the
edge connector.

Connection



Signal Conditioning 1.4

Verstärkerschaltung Gain: $G_{amp} = 2$

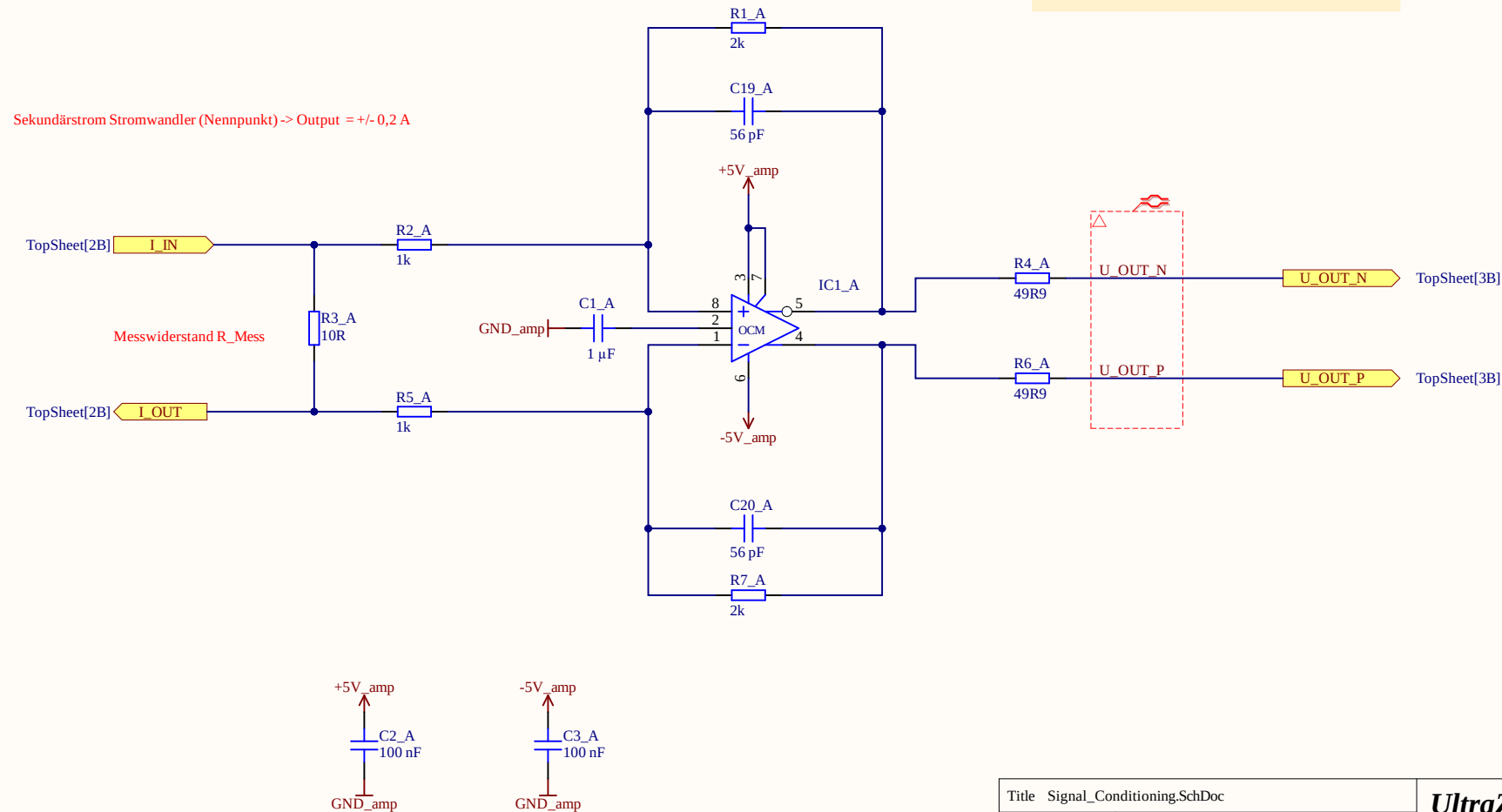
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g

Rückkopplungskondensator: $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

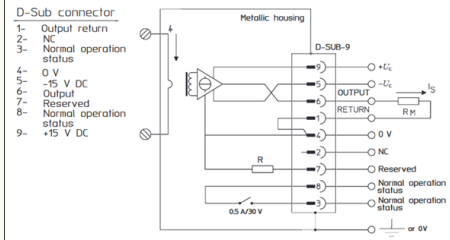
www.ultrazohm.com

Date: 06/05/2026

Sheet 2.1 of 12



Connection



Signal Conditioning 1.4

Verstärkerschaltung Gain: $G_{amp} = 2$

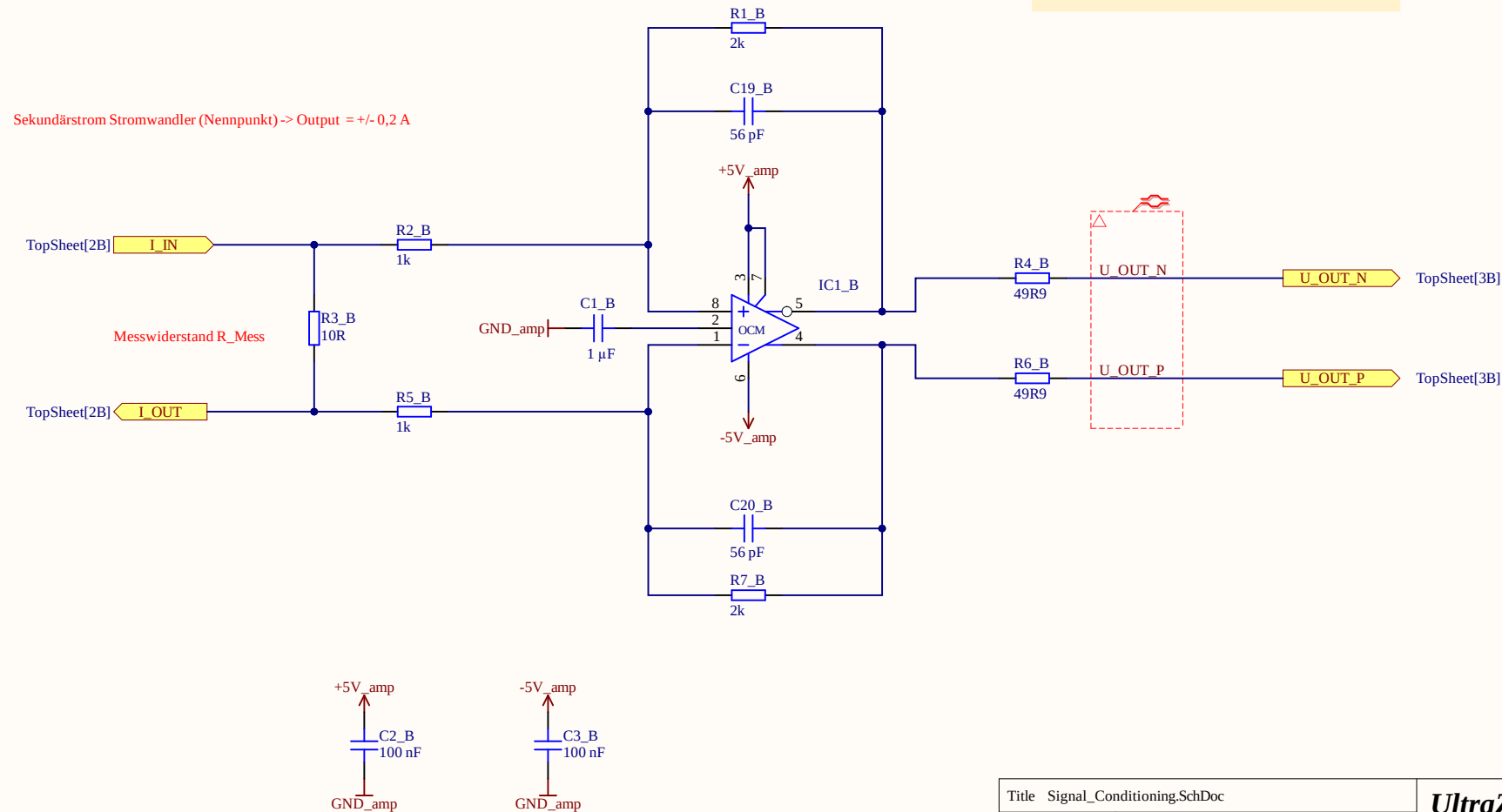
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g

Rückkopplungskondensator: $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

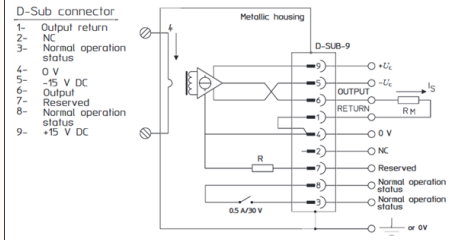
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Date: 06/05/2026

Sheet 2.2 of 12



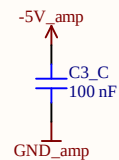
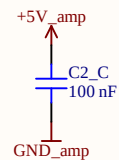
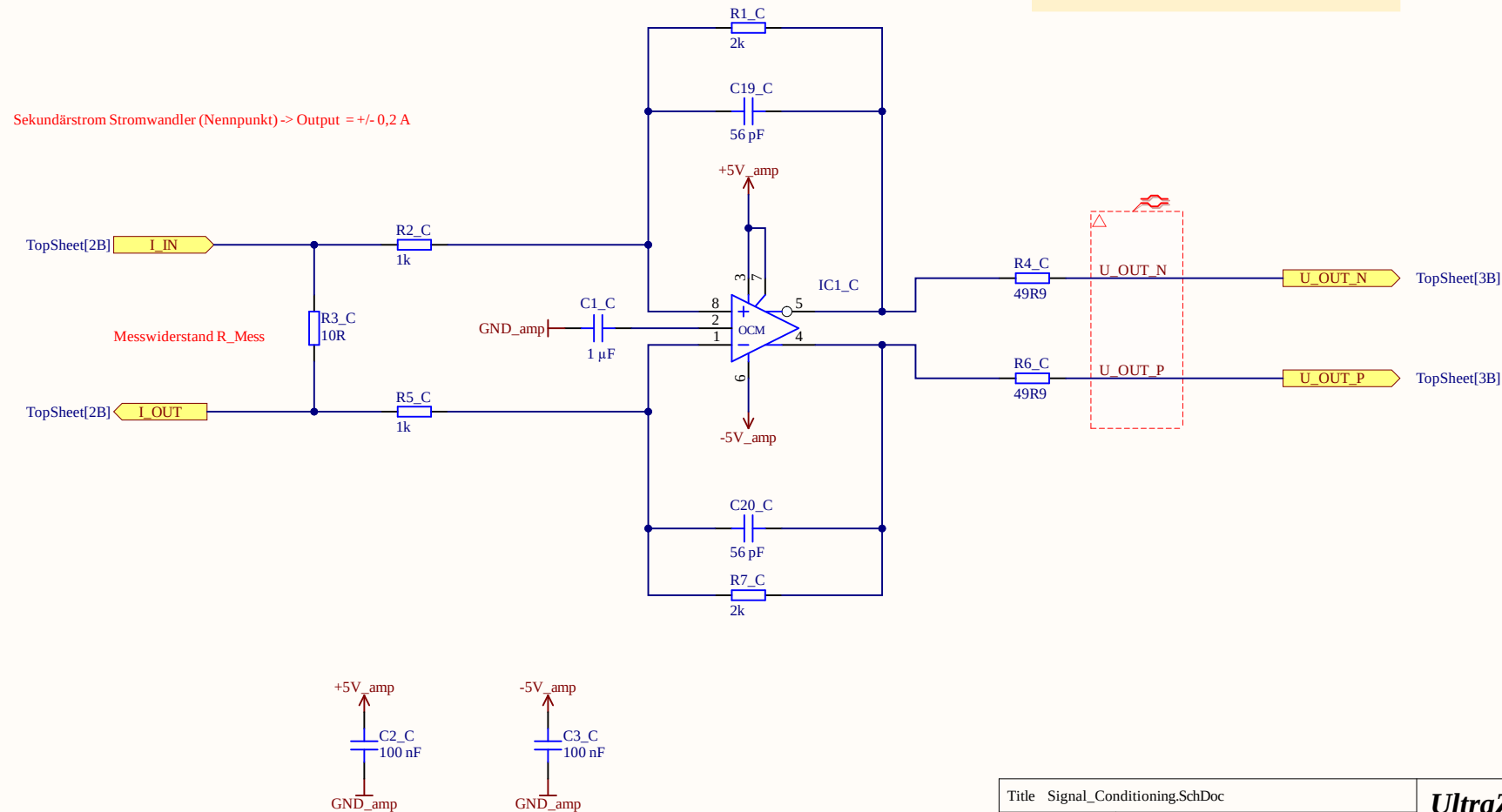
Connection



Signal Conditioning 1..4

Verstärkerschaltung Gain: $G_{amp} = 2$ $G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g Rückkopplungskondensator: $C_f =$ $C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$ 

Title Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

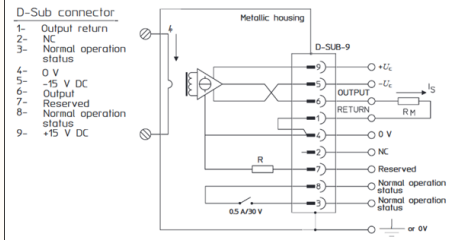
www.ultrazohm.com

Date: 06/05/2026

Sheet 2.3 of 12



Connection



Signal Conditioning 1.4

Verstärkerschaltung Gain: $G_{amp} = 2$

$G_{amp} = R_F / R_G$

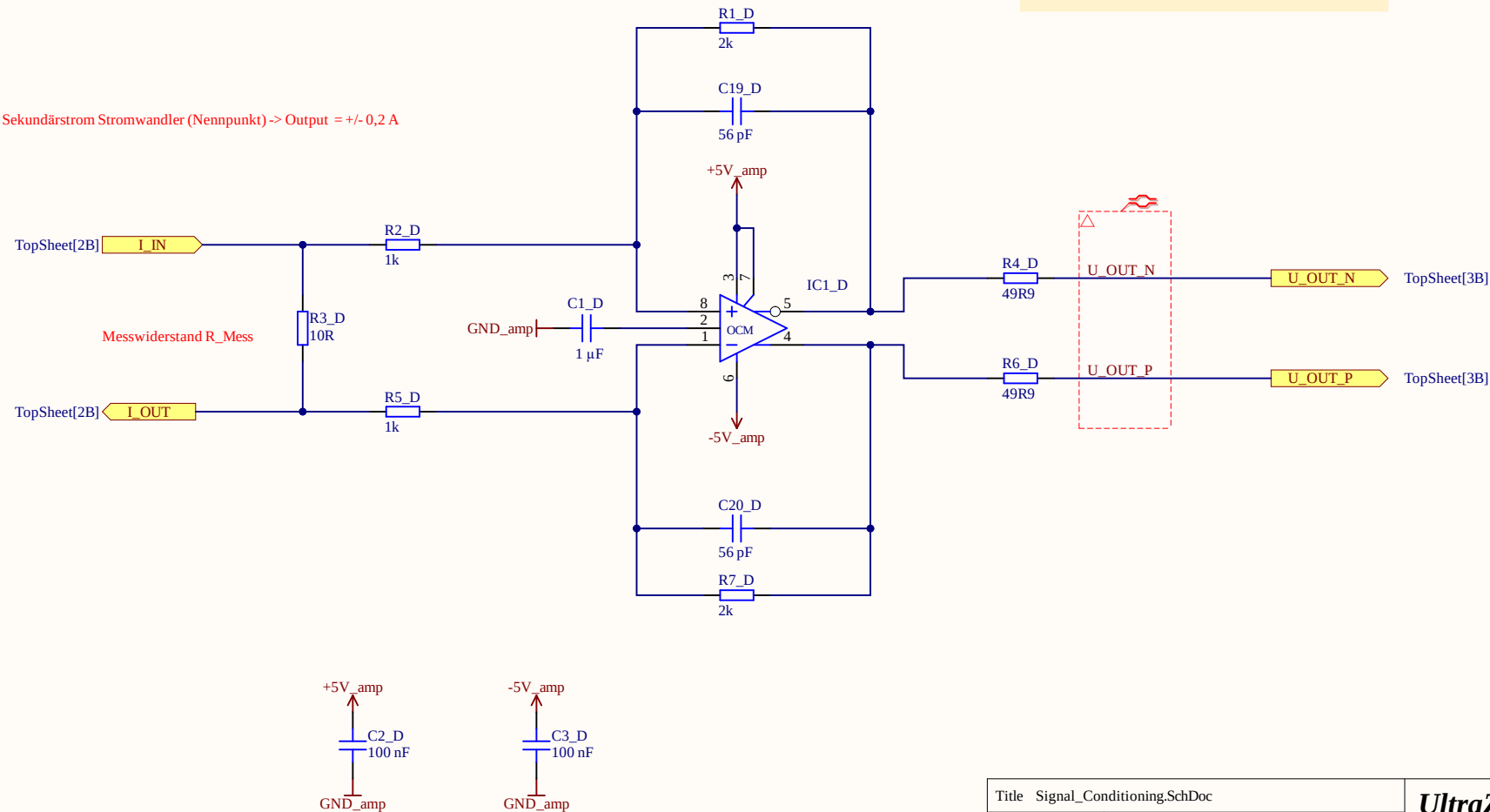
Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g

Rückkopplungskondensator: $C_f =$

$C_f = 1 / (2 \cdot \pi \cdot R_f \cdot f_g) = \dots =$

Sekundärstrom Stromwandler (Nennpunkt) $\rightarrow$ Output = $\pm 0,2$ A



Title Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

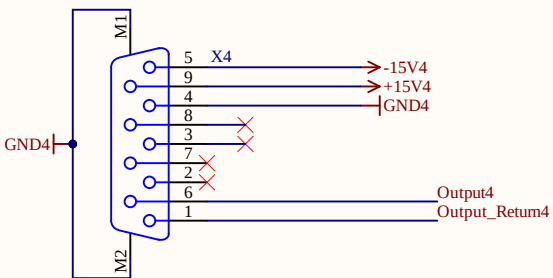
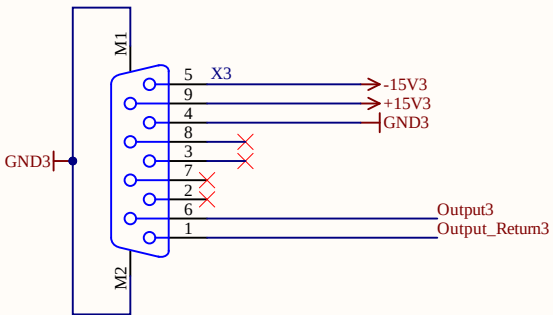
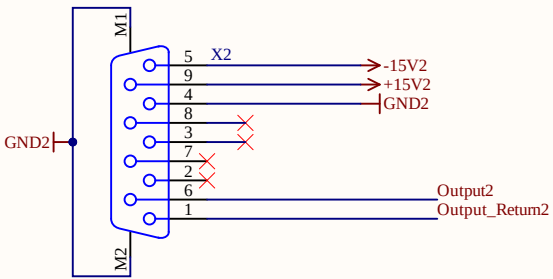
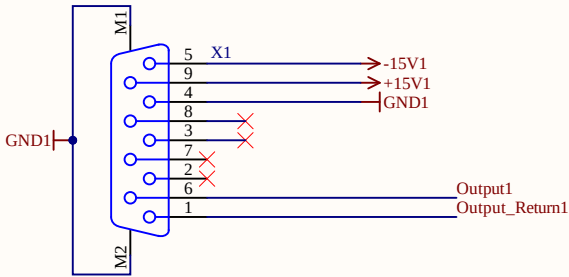
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Date: 06/05/2026

Sheet 2.4 of 12



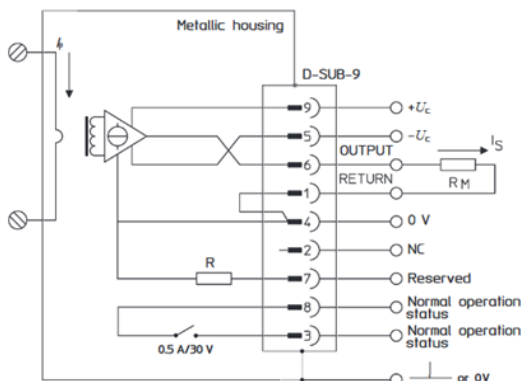
Analog Input



Connection

D-Sub connector

- 1- Output return
- 2- NC
- 3- Normal operation status
- 4- 0 V
- 5- -15 V DC
- 6- Output
- 7- Reserved
- 8- Normal operation status
- 9- +15 V DC



Output[1..4] → Output[1..4] TopSheet[1B]

Output_Return[1..4] → Output_Return[1..4] TopSheet[1B]

+15V[1..4] → +15V[1..4] TopSheet[1B]

-15V[1..4] → -15V[1..4] TopSheet[1C]

GND[1..4] → GND[1..4] TopSheet[1C]

Title DSub_Analog_Input.SchDoc

Revision: 01 Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

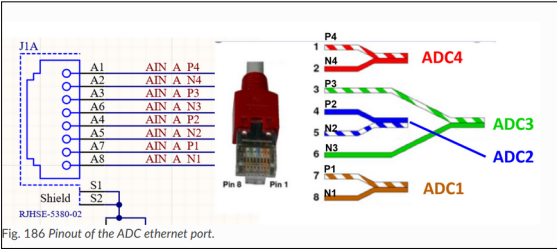
www.ultrazohm.com

Date: 06/05/2026

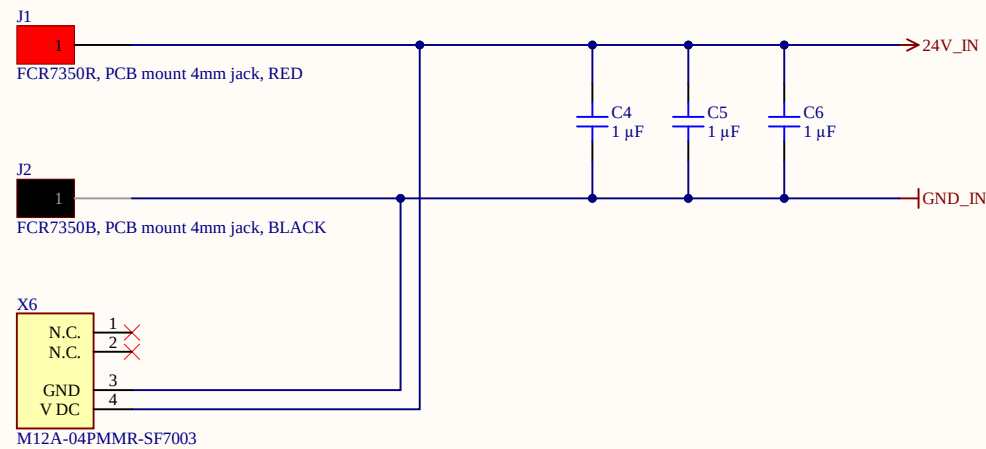
Sheet 3 of 12



Analog Output



Power Supply for Amplification Circuit



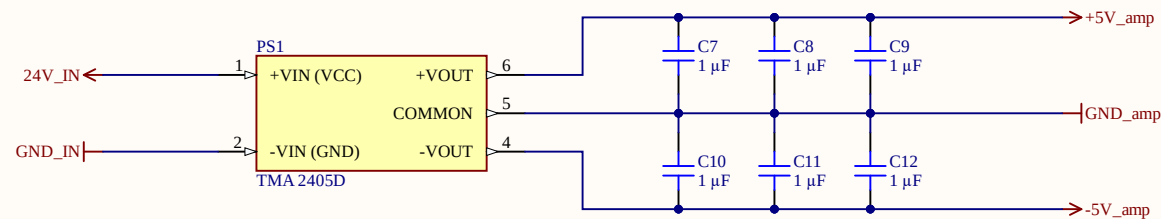
Quelle: DC/DC Wandler 24V auf +/- 5V

Verbraucher: ADA4940 fully differential Amplifier

Ruhestrom: 1,25 mA
bei 5,0 V
Power: 0,00625 mW

Output Characteristics:

$U_{out} \sim V_s = 5 \text{ V}$
 $I_{out} = 46 \text{ mA peak}$
 $P_{out max} = 0,23 \text{ W}$



Title Power_Supply_Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

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Date: 06/05/2026

Sheet 5 of 12



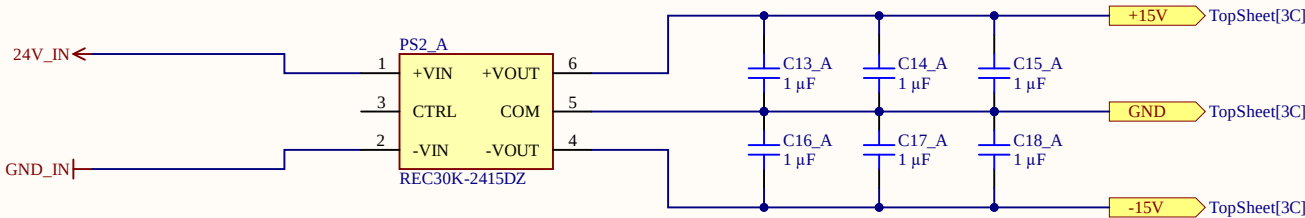
Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V
DC Nennstromaufnahmen @15V = 320 mA
I_max = 420 mA

Secondary nominal RMS current (@100A) = 0,2 A
Shielded output cable & shielded plug are recommended to use

Max Power Consumption (pro Stromsensor):
P = U*I_max = 15 V * 0,420 A = 6,3 W



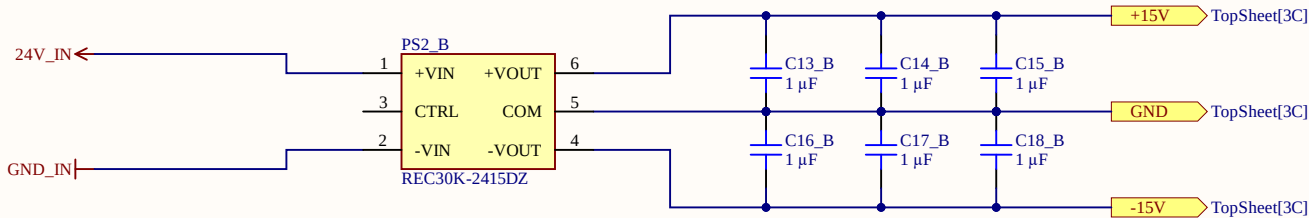
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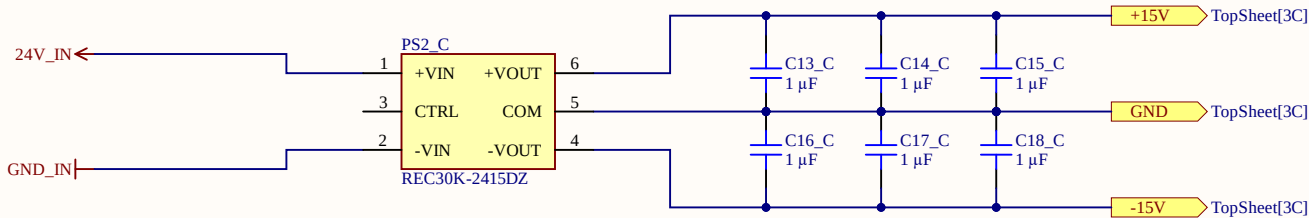
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